FROM UNIVERSAL PREDICTION TO UNIVERSAL WORKFLOWS: A SURVEY OF FOUNDATION MODELS FOR TABULAR AND TIME SERIES DATA

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ABSTRACT

Structured data plays a central role in many critical applications, ranging from finance to healthcare. While Foundation Models have achieved remarkable success in natural language processing and computer vision, extending them to structured data—primarily tabular and time series—remains challenging. Although tabular and time series data exhibit different structural properties, they share several fundamental difficulties, including heterogeneous data schemas, the lack of a fixed vocabulary for continuous values, and substantial distribution shifts across datasets. In response to these challenges, both communities have developed closely related ideas and solutions, despite using different terms and modeling approaches. This survey brings together recent advances in Foundation Models for time series and tabular data under a unified perspective. We systematically organize existing methods along four key dimensions: *Input Encoding*, *Model Architecture*, *Pre-training Strategies*, and *Adaptation Techniques*, covering both native structured-data foundation models and LLM-adapted approaches. Beyond task-level prediction, we further extend the discussion to process-level generalization by reviewing emerging agentic frameworks that support end-to-end data workflows. Finally, we outline several promising research directions, including omni-modal intelligence and composable ecosystems. A comprehensive list of papers is available at https://github.com/LAMDA-Tabular/Awesome-FMSD.

***K*eywords** Structured Data, Tabular Data, Time Series, Foundation Model, LLM Agents

1 Introduction

In the era of data-driven decision-making, structured data remains the cornerstone of information storage and analysis across many domains, ranging from financial risk assessment and healthcare monitoring to the industrial Internet of Things. Unlike unstructured modalities such as text or images, structured data is characterized by its explicit schema, and is most commonly studied in two representative and closely related modalities: *Time Series* and *Tabular Data*.

As illustrated in Figure 1, *Time Series* focus on modeling *temporal dependencies* across numerical sequences indexed by timestamps, whereas *Tabular Data* targets *feature interactions* among heterogeneous features across instance rows. However, distinct from the traditional views that treat them as isolated modalities, they share a strong structural connection. In practice, time series are frequently archived in tabular schemas—stored as *Longitudinal Panel Data*—where rows correspond to timestamps and columns to multivariate channels. Correspondingly, tabular datasets often implicitly represent *Temporal Snapshots* of dynamic systems (*e.g.*, a patient's vitals at admission), where each row is a discrete slice of a continuous timeline [1, 2, 3]. This structural overlap challenges the strict *i.i.d.* assumption traditionally applied to tabular data, revealing that the boundary between time series and tables can sometimes be characterized as a matter of indexing rather than a fundamental divergence in information content [4].

Driven by the need to capture these complex dynamics and scale beyond single-dataset limitations, the modeling paradigm has gone through several stages over time. *Early work* mainly relied on traditional statistical and tree-based methods—such as ARIMA for linearity in time series [5] and XGBoost for decision boundaries in tables [6]—which, while effective, lacked transferability. *Later*, *deep multi-task models emerged*, aiming to handle multiple tasks within

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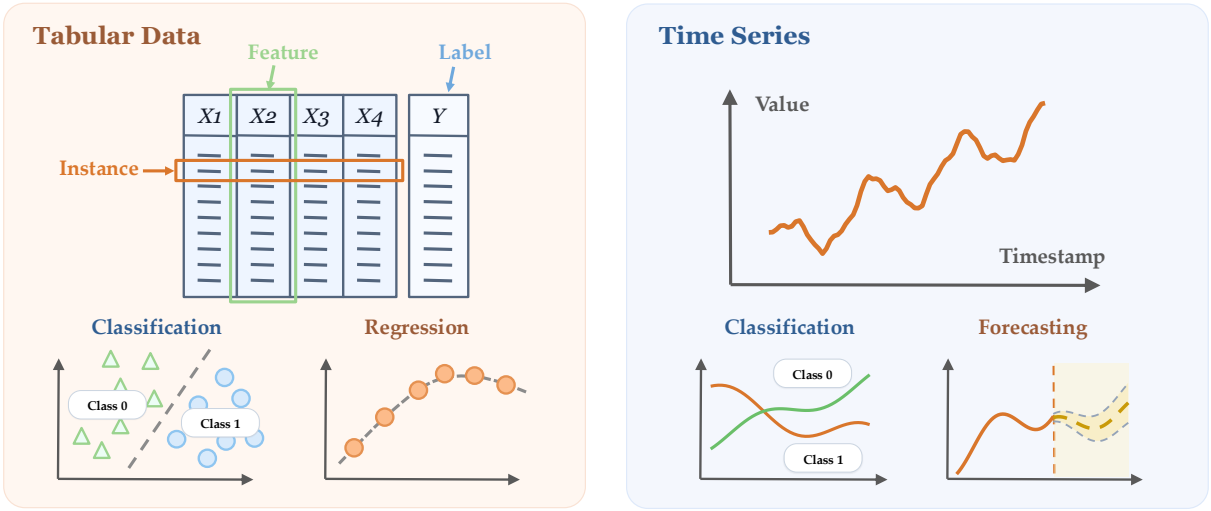


Figure 1: Examples of structured data. **Left:** Tabular data, structured as matrices of instance rows and heterogeneous feature columns, primarily targets classification and regression tasks based on feature interactions. **Right:** Time series, indexed by timestamps, focuses on capturing temporal dependencies for tasks such as classification and forecasting over univariate or multivariate sequences.

shared model architectures. In the time series domain, architectures like TimesNet [7] introduced 2D variation modeling to handle forecasting and anomaly detection within a single backbone. Similarly, for tabular data, TransTab [8] utilized contrastive learning to enable transfer learning across tables with distinct schemas. *More recently*, the rapid advancement of Large Language Models (LLMs) led to the emergence of *LLM-Adapted Models*, such as Time-LLM [9] and TabLLM [10], which align numerical inputs with pre-trained language models through reprogramming or serialization, respectively, to leverage pre-trained linguistic knowledge, effectively bridging the *modality gap*.

In recent years, both time series and tabular research have moved toward *native foundation models* for structured data [11]. Although these models come from different research communities, they are motivated by similar challenges that arise in structured data. These challenges include *schema heterogeneity* (inconsistent feature dimensions), the *tokenization bottleneck* (continuous values are incompatible with discrete vocabularies), and *distribution diversity* across datasets. Native foundation models address these challenges using similar design ideas. For time series, *time series foundation models* such as Moirai [12] rely on large-scale real-world data and patch-based representations to learn general temporal patterns. For tabular data, *tabular foundation models* like TabPFN [13] are trained on synthetic data to support in-context learning across diverse schemas without additional parameter updates.

While prior work has separately studied tabular learning [14, 15, 16, 17] and time-series learning [18], these efforts often treat the two domains in isolation or focus primarily on domain-specific modeling paradigms. They therefore leave the structural similarities between tabular and time series data underexplored, especially from the perspective of foundation models. To fill this gap, this survey presents a unified view of *foundation models for structured data*, focusing on time series and tabular data.[2] Rather than treating these domains separately, we introduce a unified taxonomy that organizes existing methods along key dimensions, including input encoding, model architecture, pre-training, and adaptation.

Here, structured-data foundation models are pre-trained across datasets or task distributions and reused on unseen datasets through zero-shot or in-context inference, transferable representations, or lightweight adaptation. We focus on native TFMs/TSFMs, include cross-modal predictors at the model level and structured-data agents at the workflow level, exclude single-dataset models and generic agents without a structured-data focus, and cover public work through September 2026. This scope traces the progression from universal prediction to universal workflows. The contributions of this survey are summarized as follows:

- We propose a *unified taxonomy* that bridges Time Series and Tabular Foundation Models, organizing existing methods along key dimensions such as input encoding, architecture, pre-training, and adaptation.
- We extend the discussion from model-level *universal prediction* to workflow-level generalization by reviewing *agentic systems* that automate predictive pipelines and reason over structured data.
- We outline open directions for structured foundation models, including *omni-modal intelligence* and *composable ecosystems*.

[2]While acknowledging diverse tasks (*e.g.*, anomaly detection), this survey centers on core predictive modeling for the two dominant structured modalities: *Time Series* and *Tabular Data*.

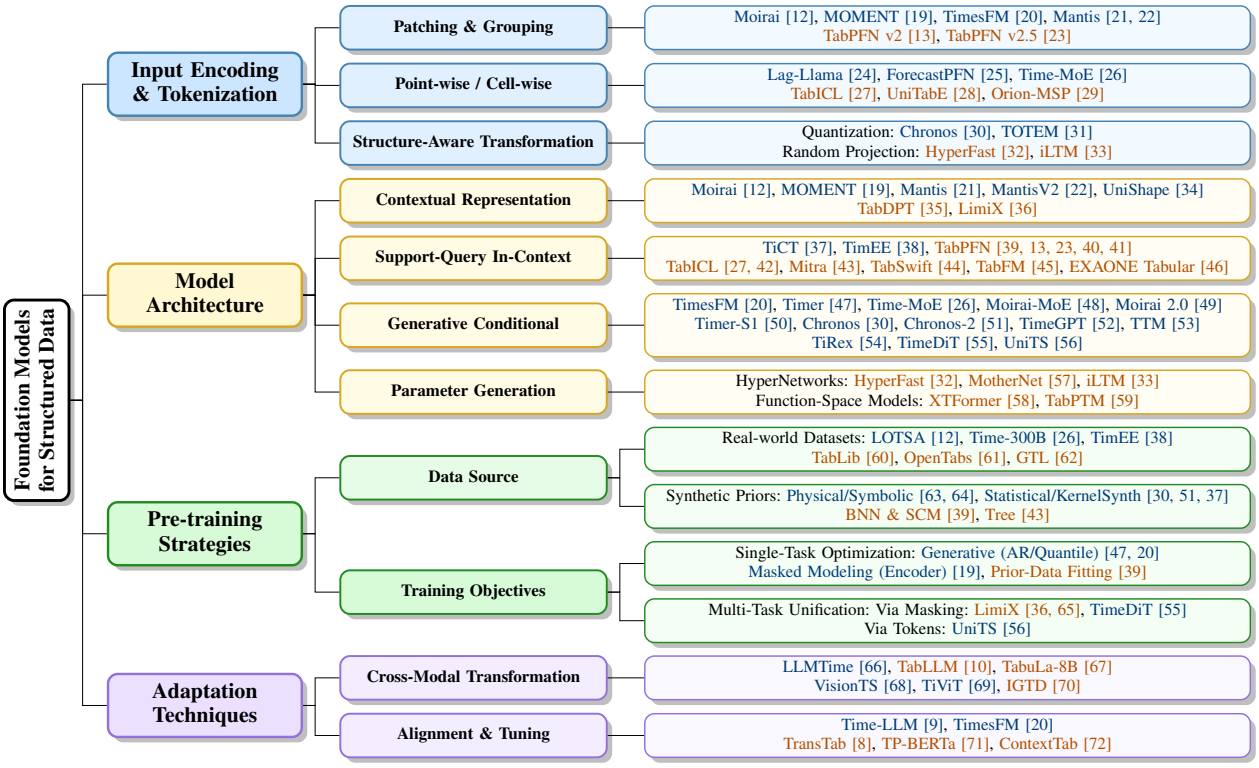


Figure 2: A unified taxonomy for structured data foundation models. **Blue** denotes **Time Series** methods, and **Orange** denotes **Tabular** methods. Methods are representative rather than exhaustive and are placed according to their primary role.

2 Preliminaries

We formulate the learning problem for structured data, identifying the fundamental differences from unstructured modalities and proposing a unified framework for Foundation Models.

2.1 Problem Formulation: Tabular and Time Series

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ denote a dataset drawn from a distribution $\mathcal{P}$. The goal is to learn a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ that minimizes the expected risk $\mathbb{E}_{(\mathbf{x},y)\sim\mathcal{P}}[\ell(f_\theta(\mathbf{x}), y)]$. Despite this shared goal, the two data types impose distinct structural characteristics.

A tabular dataset consists of N samples with d heterogeneous features. Unlike sequential data, tabular data is often modeled under *Dual Permutation Invariance*. At the *instance level*, assuming exchangeable samples, the joint likelihood is invariant to row reordering. At the *feature level*, a sample $\mathbf{x} \in \mathbb{R}^d$ represents a set of attributes rather than a sequence; hence, predictions should remain unchanged when feature values and their associated identities are permuted jointly. Consequently, models must capture complex feature interactions $x^{(j)} \leftrightarrow x^{(k)}$ without relying on positional information typically used in NLP or CV.

In contrast, a time series input $\mathbf{x} \in \mathbb{R}^{L\times V}$ represents a sequence of length L with V variates. For forecasting, the target y typically corresponds to the future horizon $\mathbf{x}_{L+1:L+H}$. For classification, y denotes a series-level categorical label. Therefore, time series foundation models may aim either to model future temporal dynamics or to extract discriminative temporal evidence for label inference. Structurally, time series are defined by dependencies along two axes. Along the *Temporal Axis*, observations often exhibit serial dependence. Forecasting conditions future targets on past observations, and autoregressive models commonly enforce this direction through causal masking; in contrast, classification, representation learning, and imputation may use bidirectional context over the observed sequence. The *Spatial (Variate) Axis* presents a modeling trade-off: while variates often exhibit inter-channel correlations similar to tabular features, forecasting-oriented time series models frequently adopt *Channel Independence*—treating multivariate series as independent univariate tokens—to mitigate noise and distribution shifts across heterogeneous channels [73].

2.2 Structural Modality *vs.* Unstructured LLMs

While the Foundation Model paradigm has revolutionized NLP and CV, extending its capabilities to structured data is challenging due to conflicting inductive biases, creating inherent "Modality Gaps." On the one hand, regarding structural dependencies, LLMs rely on sequential ordering. In contrast, tabular features lack an intrinsic positional order, so models should handle joint permutations of feature values and identities without relying on arbitrary column positions. On the other hand, regarding data representation, a fundamental mismatch exists between the finite, discrete vocabulary $\mathcal{V}$ of LLMs and the continuous numerical values ($\mathbb{R}$) of structured data. Bridging this "Tokenization Bottleneck"—mapping continuous values to discrete tokens while preserving task-relevant information and adequate numerical fidelity—remains a key challenge.

2.3 Unified Challenges and Foundation Models

A Foundation Model (FM) for structured data aims to learn a general-purpose parameter set θ^* capable of generalizing across a distribution of heterogeneous tasks $\mathcal{T}$. We formulate this as minimizing the risk over tasks $\tau \sim \mathcal{T}$, conditioned on a context $\mathcal{C}_\tau$ (*e.g.*, historical data or support set):

$$\theta^* = \arg\min_{\theta} \mathbb{E}_{\tau \sim \mathcal{T}} \left[\sum_{(\mathbf{x},y)\in\mathcal{D}_\tau} \ell\left(f_\theta(\Phi_\tau(\mathbf{x}), \mathcal{C}_\tau), y\right) \right]. \tag{1}$$

This formulation highlights key shared fundamental challenges. The first involves *Heterogeneity and Vocabulary Mismatch*. Since tasks vary in feature dimension d_τ and underlying semantics (*e.g.*, "Voltage" *vs.* "Age"), current methods must employ a flexible encoder Φ_τ—such as *Patching* in TSFMs or *Feature Tokenizers* in TFMs—to project diverse input spaces into a unified latent dimension. Furthermore, to handle continuous y and $\mathbf{x}$, Φ_τ often integrates *Serialization* or *Native Numerical Embeddings* to address the lack of a fixed vocabulary. Another main challenge is *Distribution Diversity*. Structured data exhibits significant distribution shifts across domains. FMs address this via the context term $\mathcal{C}_\tau$. By employing *In-Context Learning*, the model approximates the posterior $P(y|\mathbf{x}, \mathcal{C}_\tau)$ at inference time, enabling zero-shot adaptation to new distributions without parameter updates.

3 Taxonomy

We propose a unified taxonomy for Foundation Models (FMs) in structured data. Rather than separating time series and tabular models into two isolated surveys [18, 14], we organize existing methods by shared design questions: how heterogeneous structured inputs are encoded, how the model interfaces with prediction, how pre-training induces cross-dataset generalization, and how pre-trained models are adapted to new tasks. Under each dimension, we first identify the common challenge and then contrast the modality-specific inductive biases. This design-oriented view highlights why similar solutions often emerge independently across the two communities.

3.1 Input Encoding and Tokenization

The projection of heterogeneous structured values into model-compatible latent representations constitutes the *Tokenization Bottleneck*. Unlike NLP's fixed vocabulary, structured data consists of heterogeneous values with dynamic distributions. Current research addresses this through three paradigms: *Patch-wise / Group-wise Encoding* (focusing on information density and computational efficiency), *Point-wise / Cell-wise Encoding* (focusing on fine-grained precision), and *Structure-Aware Transformation* (designed to preserve modality-specific constraints).

Aggregation via Patching and Grouping. To improve *Information Density* and reduce the quadratic complexity of attention, aggregation compresses signals into semantic units. In *Time Series*, *Patch-wise Tokenization* segments signals into sub-sequences. While standard models (*e.g.*, **TimesFM** [20]) use fixed-length patching, recent *Dynamic Patching* approaches (*e.g.*, **LPTM** [74], **Kairos** [75]) leverage entropy-aware segmentation, adaptively allocating higher resolution to fluctuating regions while compressing stable trends. Beyond *forecasting-oriented* models, patch-wise tokenization is also adopted by *classification-oriented* TSFMs. **Mantis** [21] and **MantisV2** [22] use patch-based ViT-style encoders to transform raw sequences into transferable embeddings for downstream time series classification, showing that patching can serve not only generative forecasting but also discriminative representation learning. A parallel trend occurs in *Tabular Data*, shifting from fixed *Global Projection* (*e.g.*, the original **TabPFN** [39] and **TabDPT** [35]) to scalable *Stochastic Grouping*. Newer architectures such as **TabPFN v2** [13] and **TabPFN v2.5** [23] jointly embed small feature groups, with the latter increasing group size to improve training and inference efficiency. Their alternating feature- and sample-axis attention then models dependencies across these groups. **TabPFN-3** [40] adopts the cyclic triplet grouping of **TabICLv2** [42], linearly projecting grouped values with missing/infinite-value indicators.

Fine-grained Tokenization for Atomic Fidelity. In contrast, another approach prioritizes *Fine-grained Representation*, mapping fundamental data units directly to tokens to preserve detailed information. In *Time Series*, *Point-wise Tokenization* avoids segmentation. Models like **Lag-Llama** [24] and **Time-MoE** [26] embed individual time steps or lagged windows, retaining complete temporal resolution at the cost of shorter context windows. In tabular domains, *Cell-level Tokenization* treats each feature-value pair as an individual token. Architectures such as **TabICL** [27] and **Mitra** [43] employ this strategy to model explicit inter-column interactions via attention, compressing the context into a specific summary token for prediction. This granular approach is critical for tasks requiring precise feature attribution.

Structure-Aware Transformation. Beyond generic tokenization, specialized strategies employ domain-specific transformations to address the modality gap. For time series, *Discrete Value Quantization* aligns continuous signals with the discrete priors of Language Models. **Chronos** [30] quantizes values into bins to reframe forecasting as classification, while **TOTEM** [31] learns a VQ-VAE codebook to represent waveform shapes as tokens. Similarly, for tabular data, discretization mechanisms such as **TP-BERTa** [71] leverage decision trees to partition continuous features into discrete leaves, functioning similarly to time series quantization. Beyond value-level transformation, strategies also address *Schema Unification*. To handle high heterogeneity, **HyperFast** [32] and **iLTM** [33] utilize *Random Projection* to map arbitrary input dimensions into a fixed-size latent space, effectively converting diverse schemas into a unified input format. **CoCoTab** [76] uses row and column patches to capture the structural relationships within tables. Conversely, **TabPFN-TS** [77] repackages time series forecasting as tabular regression through lightweight temporal featurization, allowing a tabular foundation model to perform zero-shot forecasting.

Discussion. The evolution of input encoding reveals a clear structural parallel between modalities. *Patching* in time series and *Feature Grouping* in tabular data both serve as compression mechanisms. A key difference persists: forecasting-oriented tokenization must preserve *temporal order* and prevent future leakage, whereas classification-oriented encoders may use bidirectional temporal context. Tabular tokenization, meanwhile, should avoid dependence on arbitrary feature positions while preserving feature identities. Accordingly, patch or group aggregation is preferable for long or high-dimensional inputs, whereas point- or cell-level encoding is preferable when local numerical detail or feature-level attribution outweighs context-length cost. Structure-aware transformations are orthogonal to token granularity and can complement either choice when temporal order, feature identities, or cross-modal compatibility must be preserved. Future research suggests a move towards *Unified Statistical/Semantic Adaptive Tokenization*, where the granularity of patching or grouping dynamically adjusts to the local information content of the data, optimizing efficiency while preserving key information across both modalities.

3.2 Model Architecture

Driven by the need to model variable-length contexts and heterogeneous dependencies, transformer-based architectures have become the dominant backbone for structured foundation models. However, the key distinction is not simply whether a model serves forecasting, classification, or regression. More fundamentally, architectures differ in their *prediction interface*: how they construct context, enforce structural constraints, and produce the target output. We therefore categorize model architectures into four paradigms that cut across task boundaries.

Contextual Representation Backbones. This paradigm first maps structured inputs into transferable latent representations, which are then consumed by a task head, a probe, or a downstream classifier. For time series, masked or encoder-style models such as **Moirai** [12] and **MOMENT** [19] encode historical windows or patches while preserving temporal order through positional mechanisms such as RoPE [78] or patch-level structure. Classification-oriented TSFMs follow the same representation-first interface: **Mantis** [21], **MantisV2** [22], and **UniShape** [34] use pre-trained encoders as frozen or lightly tuned feature extractors for label prediction. In tabular modeling, shared-encoder approaches such as **TabDPT** [35] and **LimiX** [36] map heterogeneous tables into a common representation space before routing predictions to lightweight heads. Across modalities, this interface separates representation learning from task execution, making it naturally compatible with probing, transfer, and multi-task heads.

Support-Query In-Context Predictors. A second paradigm treats prediction as conditional inference over a labeled context set. Given a support set $\mathcal{C} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ and a query $\mathbf{x}_q$, the model directly estimates $p(y_q \mid \mathbf{x}_q, \mathcal{C})$ in a single forward pass. This interface is central to tabular foundation models such as **TabPFN** [39, 13, 23, 40, 41], **TabICL** [27, 42], **Mitra** [43, 79], and **TabSwift** [44]. The same support-query interface also appears in time series classification, where **TiCT** [37] and **TimEE** [38] condition on labeled support series and predict labels for query series without task-specific parameter updates; recent analysis such as **TIC-FM** [80] further argues for this protocol as a training-free evaluation interface. Architecturally, these models require permutation-aware handling of support rows and feature/label identities: **TabPFN v2** [13] uses randomized token identifiers [81], **EquiTabPFN** [82] enforces target-permutation equivariance, and mechanistic analysis links invariance and failure modes to specific positional and readout mechanisms [83]. Efficiency is another central pressure point, motivating designs such as row-wise attention and early exit in **TabSwift** [44].

Recent designs retain this interface while changing its internal computation: **TabFM** [45] substantially scales up both model size and synthetic pre-training data and incorporates learned Fourier cell embeddings, a design subsequently adopted by **TabPFN-3.5** [41]; **EXAONE Tabular** [46] interleaves feature- and support-conditioned item-axis attention; and **Xiaomi-TabLDM** [84] combines dual-stream feature grouping with sparse mixture-of-experts layers. Retaining **TabPFN-3** [40]'s grouping and missing/infinite-value indicators, TabPFN-3.5 combines learned Fourier features of standardized values with context-derived empirical cumulative distribution function encodings to capture both numerical magnitude and distribution-relative information. TabFM and Xiaomi-TabLDM additionally use held-out or out-of-fold predictions for validation-guided selective ensembling, fitting nonnegative ensemble weights without updating the pretrained model. **Causilo** [85] refines cell representations through additional row and column interactions before row compression, using fixed-size summary tokens to make within-row attention scale linearly with the number of features. **Retro** [86] uses Attention Residuals to selectively reuse intermediate representations across model depth; **Kumo Tabular** [87] combines column and row attention with row compression and cacheable support–query inference.

Generative Conditional Predictors. Generative architectures model the target as a structured continuation or reconstruction conditioned on observed context. This paradigm is most visible in time series forecasting, where decoder-only models such as **TimesFM** [20], **Timer** [47], **Time-MoE** [26], **Moirai-MoE** [48], **Moirai 2.0** [49], and **Timer-S1** [50] formulate prediction as next-patch or next-token generation. Quantization-based models such as **Chronos** [30] convert continuous values into discrete token sequences. **Chronos-2** [51] instead uses continuous patch embeddings, group attention, and multi-patch quantile outputs. Other generative predictors follow different architectural routes: **TimeGPT** [52] uses a Transformer encoder–decoder; **TTM** [53] combines a lightweight TSMixer backbone with a compact decoder and linear forecast head; and **TiRex** [54] adopts a decoder-only recurrent xLSTM architecture for patch-wise forecasting. Multivariate extensions include **TimesFM-3** [88], which natively supports jointly evolving series and past-only or future-known covariates, and **TiRex-2** [89], which combines recurrent temporal modeling with cross-variate attention and constant per-patch inference cost under streaming. **Toto** [90] uses a decoder-only architecture tailored to multivariate observability data. Diffusion-based models such as **TimeDiT** [55] extend the same conditional-generation view to iterative denoising. Although these models are often introduced for forecasting, the interface naturally extends to imputation and reconstruction by changing the observed and target masks, which makes it a task-unifying rather than purely forecasting-specific architecture.

Parameter-Generating Architectures. The fourth paradigm does not directly output labels or future values. Instead, it maps an entire dataset context to the parameters of a task-specific predictor, $\theta = \mathcal{H}(\mathcal{D}_{\text{context}})$. This design is especially useful for tabular data, where schema heterogeneity makes fixed input-output heads difficult to share across datasets. **HyperFast** [32], **MotherNet** [57], and **iLTM** [33] generate classifier weights in a single forward pass, decoupling dataset-level adaptation from per-query inference. Related function-space approaches such as **XTFormer** [58] and **TabPTM** [59] learn reusable bases or task-specific mappings over heterogeneous tables. Compared with support-query predictors, parameter-generating models compress the context into model weights, trading direct contextual attention for fast downstream inference.

Discussion. These paradigms describe primary prediction interfaces rather than mutually exclusive task categories: classification can use representation or support–query models, tabular prediction can use in-context, shared-encoder, or parameter-generating models, and forecasting can use generative models or tabular regression [77]. Forecasting requires directionally valid conditioning to prevent future leakage, whereas representation and classification models may use bidirectional temporal context; tabular predictors instead require permutation-aware feature and label handling. Efficiency motivates alternatives such as the state-space **TimeMachine** [91], linear-attention **TabFlex** [92], permutation-aware **Hydra** [93], and lightweight in-context **TabSwift** [44]. Representation backbones suit encoder reuse; support–query predictors suit immediate adaptation from labeled context; generative predictors suit sequential, reconstructive, or distributional targets; and parameter-generating models suit repeated queries over one dataset. Their main costs are, respectively, task-specific heads or adaptation, context-length scaling, autoregressive or iterative decoding, and information loss through context compression.

3.3 Pre-training Strategies

Pre-training determines the generalization capability. We analyze this through two orthogonal dimensions: *Data Composition* (Data Sources) and *Learning Objectives* (Pre-training Tasks).

3.3.1 Data-Centric Strategies

The scarcity of standardized structured data has led to different data strategies: many tabular FMs use synthetic priors to span heterogeneous schemas, whereas many time-series FMs aggregate predominantly real-world, cross-domain corpora to learn recurring temporal patterns.

Tabular Data: The Synthetic Prior Hypothesis ($\mathcal{T}_{\text{syn}}$). Addressing the lack of universal schemas, the **PFN Family** argues that real-world data may be insufficient for broad generalization. Instead, they train on synthetic tasks $\tau \sim \mathcal{T}_{\text{syn}}$ to learn the "algorithm of inference." While **TabPFN** [39] relied on SCM- and BNN-based priors, successors enhance prior complexity: **TabForestPFN** [94] incorporates tree-based structures, and **Mitra** [43] optimizes prior selection via a "Generalizability Matrix" to cover real-world decision manifolds. **Mitra-v2** [79] retains the SCM and tree-based generators of **Mitra-v1** and adds a hybrid SCM generator to the mixture. For regression, **Nori** [95] combines SCM-based priors with diverse explicit regression function families. Conversely, *Real-World Aggregation Methods* (*e.g.*, **TabDPT** [35] and **GTL** [62]) use large collections of real tables to capture semantic co-occurrences missing from synthetic distributions.

Time Series: The Empirical Scaling Hypothesis ($\mathcal{T}_{\text{real}}$). For forecasting, the shared sequential numerical interface makes cross-domain aggregation feasible, motivating empirical scaling over large corpora. Resources such as Time-300B [26] and the trillion-scale TimeBench [96] provide broad coverage across domains and temporal patterns. **Toto** [90] mixes observability telemetry, public datasets, and synthetic series; **Toto 2.0** [97] further demonstrates consistent forecasting improvements when scaling a shared training recipe from 4M to 2.5B parameters. The emerging classification branch complicates this dichotomy between real-world scaling and synthetic priors. On the one hand, models such as **Mantis** [21] and **TimEE** [38] reflect the value of learning transferable representations or in-context classifiers from collections of labeled classification tasks. On the other hand, **TiCT** [37] and recent synthetic-data studies suggest that synthetic priors can also support time series classification [98, 99], echoing the prior-data fitting philosophy in tabular foundation models. In this sense, time series classification may become an important testbed for examining whether synthetic priors can complement or even replace large-scale real-world pre-training in structured data foundation models.

3.3.2 Training Objectives

The training objectives of Structured Foundation Models have evolved from task-specific metrics to unified multi-task objectives, enabling generalization across diverse dataset contexts.

Single-Task Optimization. Time series forecasting models commonly use autoregressive next-patch prediction. **Timer** [47] and **TimesFM** [20] predict continuous patches using regression losses. A generic objective is

$$\mathcal{L}_{\text{NTP}}(\theta) = \sum_{k=1}^{K-1} \ell(g_\theta(\mathbf{x}_{\leq k}), \mathbf{x}_{k+1}), \tag{2}$$

where ℓ may be a point or quantile regression loss. Alternatively, **TimeDiT** [55] employs *Score Matching* for iterative denoising. For classification, TSFMs learn transferable representations or directly infer query labels from labeled context. **Mantis** [21] learns general-purpose embeddings for downstream classification, while **TiCT** [37] and **TimEE** [38] optimize label prediction over contextual classification tasks. In the tabular domain, **TabPFN** [39] is trained with cross-entropy on query labels from classification tasks sampled from a synthetic prior. At the population optimum, this objective approximates the prior-induced posterior predictive distribution, but training does not explicitly minimize a KL divergence to it.

Multi-Task Unification. Unifying diverse tasks (*e.g.*, forecasting, imputation, and classification) requires a common modeling interface across heterogeneous input-output formats. One strategy is prompt- or mask-conditioned modeling. **UniTS** [56] demonstrates Context-Conditional Masked Modeling, where task prompts $\mathbf{m}_{\text{task}}$ guide a shared model to switch modes via

$$\mathcal{L}_{\text{Unified}}(\theta) = \ell_{\text{task}}(f_\theta(\mathbf{x}_{\text{context}}, \mathbf{m}_{\text{task}}), \mathbf{x}_{\text{target}}).$$

Varying $\mathbf{m}_{\text{task}}$ enables the same backbone to handle different prediction modes, such as target prediction, reconstruction, and forecasting, within a single pre-training framework. Building on **LimiX** [36], **LimiX-2** [65] scales context-conditional masked modeling that combines target prediction with masked-feature reconstruction, supporting classification, regression, and imputation within a shared model. At the architectural level, **TabDPT** [35] and **LimiX** [36] adopt a *Shared-Backbone Multi-Head* architecture. They use a universal encoder to extract generalized representations, which are then routed to lightweight task-specific heads, decoupling shared representation learning from task execution.

Discussion. A fundamental distinction exists in pre-training paradigms. Many tabular FMs use synthetic task distributions to learn task-conditioned inference, whereas many forecasting TSFMs scale pre-training over real-world observations to capture recurring temporal patterns. A promising direction lies in *Generative Curriculum Learning*. The Adversarially Pre-trained Transformer [100] uses synthetic data agents that shift their generating distributions to challenge the model, moving from static prior sampling toward active curriculum construction. Synthetic-prior training is preferable when heterogeneous schemas and limited reusable data make corpus aggregation difficult, but its effectiveness depends on prior–target alignment. Real-data scaling is preferable when large reusable corpora with sufficiently compatible sampling and semantics are available, but it raises curation, leakage, imbalance, and domain-coverage concerns. Hybrid curricula can combine empirical coverage with controlled synthetic variation.

3.4 Adaptation Techniques

Building upon pre-trained backbones, adaptation strategies are essential to specialize general capabilities for downstream tasks. This field includes two distinct paradigms: adapting *Native Structured Foundation Models* and adapting *Cross-Modal Foundation Models*. While the former focuses on domain specialization, the latter requires bridging the *Modality Gap* between numerical structured inputs and the representation spaces of pre-trained language or vision models.

3.4.1 Cross-Modal Alignment

The goal is to define a mapping $\phi : \mathcal{X}_{\text{struct}} \to \mathbb{R}^{d_{\text{model}}}$ that aligns numerical distributions with the representation spaces of external foundation models, such as linguistic or visual embeddings.

Direct serialization (*e.g.*, naive string conversion) leads to excessive token costs for high-frequency data. Consequently, *Model Reprogramming* approaches like **Time-LLM** [9] freeze the backbone and use a prototype-based multi-head cross-attention layer to map continuous patch embeddings into the LLM representation space, effectively reprogramming the LLM's pattern recognition capabilities for temporal dynamics. Similarly, **LLM4TS** [101] uses a two-stage alignment to match time series embeddings with the LLM's semantic space prior to fine-tuning.

Tabular LLMs must serialize rows while preserving feature identities and controlling how examples from the same table interact. Template-based serialization (*e.g.*, **TabLLM** [10] and **LIFT** [102]) can incur substantial token costs. **TabuLa-8B** [67] introduces *Row-Causal Tabular Masking (RCTM)*, which permits attention to preceding rows from the same table while preventing attention across unrelated tables, thereby supporting efficient packing and few-shot conditioning. Addressing the *Numerical-Semantic Discrepancy*, **TP-BERTa** [71] employs *Relative Magnitude Tokenization (RMT)*: feature-specific C4.5 binning maps continuous values to shared magnitude tokens, whose embeddings are scaled by the original values.

Beyond text-centric adaptation, an alternative approach leverages pre-trained vision models to bypass textual bottlenecks by utilizing spatial inductive biases. **VisionTS** [68] reformulates forecasting as visual inpainting by converting 1D time series into 2D heatmaps for masked autoencoders. In time series classification, **TiViT** [69] similarly converts time series into images and reuses frozen ViT/OpenCLIP representations, showing that visual foundation models can provide strong discriminative features for non-visual temporal data. For high-dimensional tabular data, **IGTD** [70] optimizes a feature-to-pixel mapping, transforming statistical dependencies into spatial proximity for efficient CNN/ViT processing.

3.4.2 Tuning Strategies

Strategies divide based on backbone flexibility: *Context Optimization* for fixed inference engines and *Parameter-Efficient Fine-Tuning (PEFT)* for adaptable models.

Context Optimization. For models encoding fixed inference algorithms (*e.g.*, **TabPFN**), weight updates may lead to *Prior Forgetting*. Adaptation thus shifts to optimizing the context $\mathcal{C}$. **TuneTables** [103] employs context distillation to synthesize a "super-context" $\mathcal{C}^*$ that maximizes target likelihood:

$$\mathcal{C}^* = \arg\max_{\mathcal{C}, |\mathcal{C}|<K} \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}}[\log P_{\text{PFN}}(y \mid \mathbf{x}, \mathcal{C})]. \tag{3}$$

Alternatively, **BETA** [104] focuses on *feature input optimization*, freezing the backbone while learning a transformation to project downstream data into the model's original context space.

Parameter-Efficient Fine-Tuning. For adaptable backbones, parameter-efficient fine-tuning (PEFT) is a common strategy for balancing task adaptation and retention of pre-trained behavior. **GPT4MTS** [105] adds modality-specific prompt layers and freezes GPT-2's attention and feed-forward blocks while tuning lightweight input, output, and normalization-related components. For native tabular FMs, recent evaluations [106] suggest that the benefits of PEFT and full fine-tuning are model- and data-dependent; full fine-tuning can also reduce accuracy or calibration quality.

Discussion. Adapting FMs to structured data presents a fundamental trade-off between *Semantic Alignment*, *Numerical Precision*, and *Inference Efficiency*. *Text Serialization* (*e.g.*, **TabLLM**) can exploit schema semantics but often incurs high token costs and may represent numerical values poorly. *Numeric Patching/Embedding* (*e.g.*, **Time-LLM**) reduces token length and retains numerical structure more directly, but requires an explicit mechanism to align the resulting embeddings with the pre-trained representation space. In practice, frozen alignment, context optimization, or PEFT is preferable under limited labels, tight compute budgets, or calibration-sensitive deployment, whereas deeper fine-tuning is justified when substantial in-domain data are available and task-specific gains outweigh the risks of overfitting, prior forgetting, and calibration loss. We suggest that a promising direction involves *Hybrid Adaptation*: leveraging LLMs for high-level reasoning (*e.g.*, identifying relevant variables from column names) while offloading numerical computation to specialized "Tools", connected via learnable interfaces. This tool-oriented interface links model-level universal

prediction to workflow-level generalization: structured foundation models provide reusable predictive primitives, while agents select and compose them within multi-step processes.

4 Agentic Workflows for Structured Data

The integration of Large Language Models (LLMs) has expanded structured-data analysis from static *predictors* mapping $x \rightarrow y$ toward *agents* that plan and execute multi-step workflows across the full data-science lifecycle. In this framework, the LLM acts as a central *cognitive engine*, leveraging generalized world knowledge to automate tasks ranging from semantic inspection to rationale-driven model selection. We categorize these agents by their *Scope of Autonomy* (Figure 3): *Task-Specific Specialists* act as modular experts at distinct pipeline stages, while *End-to-End Generalists* manage complete multi-step workflows. These workflows may automate predictive pipelines or support structured-data analysis and reasoning; native TFMs and TSFMs can serve as predictive primitives, although many current systems still rely mainly on general LLMs and conventional tools.

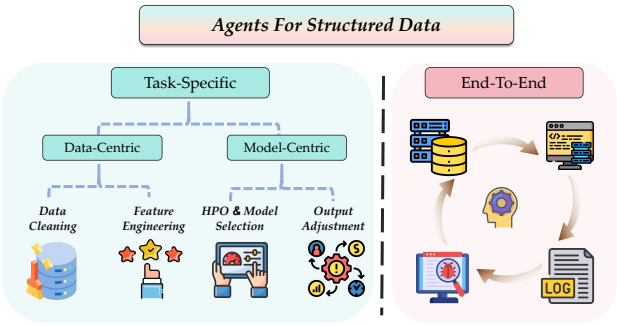


Figure 3: A taxonomy of agents for structured data, organizing them by scope of autonomy into *Task-Specific Specialists* and *End-to-End Generalists*.

4.1 Task-Specific Agents

These agents function as modular reasoning units, applying semantic logic to address challenges where statistical heuristics prove insufficient.

4.1.1 Data-Centric Agents

Data-centric agents leverage semantic reasoning to improve input quality, connecting raw storage and statistical interpretability.

Semantic Data Refinement. Data Cleaning Agents [107, 108] deploy reasoning engines to identify contextual anomalies (*e.g.*, "Age: 5, Profession: Surgeon") beyond the scope of statistical outlier detection. Unlike static rule-based systems, they perform *adaptive imputation*, generating context-aware cleaning policies that respect global dataset logic.

Semantic Feature Construction. This domain acts as a "Feature Inventor" via two streams. *Knowledge-Driven Generation* (*e.g.*, **CAAFE** [109], **ReaGen** [110]) leverages metadata within prompts to propose candidate features based on real-world logic rather than statistical correlation. Conversely, *Search-Driven Refinement* (*e.g.*, **LLM-FE** [111], **OCTree** [112]) integrates generative proposals with evolutionary algorithms or tree-based search to optimize subspace refinement. To reduce latency, inference-efficient designs like **FeatLLM** [113] decouple reasoning from execution by generating static transformation scripts, avoiding LLM calls at inference time.

4.1.2 Model-Centric Agents

These agents optimize the learner itself, applying reasoning to the dynamics of training and inference.

Reasoning-Aware HPO. Agents like **AgentHPO** [114] shift from black-box optimization to *White-Box Reasoning*. By interpreting training logs as *diagnostic traces*, agents identify optimization patterns (*e.g.*, "oscillation via high learning rate") to propose targeted configurations. Furthermore, *Meta-Learning Agents* leverage internalized priors to warm-start the search, significantly reducing the trial budget compared to random search.

Dynamic Inference Refinement. At inference, agents act as meta-reasoners. **CoT^2** [115] utilizes Chain-of-Thought to dynamically weight ensemble votes based on instance characteristics. Similarly, **DeLTa** [116] performs post-hoc refinement, optimizing decision tree rules to enhance predictive performance.

4.2 End-to-End Generalists

End-to-End agents manage complete workflows for predictive automation or structured-data reasoning.

Tree-of-Thought Planning. Advanced generalists model engineering as a search over a solution space. **AIDE** [117] constructs a *Solution Space Tree* to explore pipeline branches and backtrack from failures. To manage long horizons, **MLMaster** [118] implements an *adaptive memory mechanism*, maintaining structured memory of experiments to prevent context collapse. **TabTracer** [119] coordinates multi-step tool calls over intermediate table states, using Monte Carlo Tree Search, explicit state tracking, and rollback to support verifiable reasoning.

Workflow Automation and Structured-Data Reasoning. Some systems primarily automate predictive model development and evaluation. Reliability is the essential requirement of autonomy. **AutoKaggle** [120] implements a *Multi-Agent System (MAS)* with specialized roles (*e.g.*, Planner, Developer, Reviewer), incorporating *Test-Driven Development* for executability. **MLZero** [121] aims to achieve "Zero-Intervention," employing a perception module for raw file structures and Dual Memory for multimodal tasks. **AutoML-GPT** [122] uses GPT to generate task-oriented prompts from user requests and model/data cards, automating end-to-end training. **TabFM-Auto** [123] directly connects native foundation models to workflow automation: an LLM agent uses dataset metadata and validation feedback to refine data cleaning, feature engineering, context selection, and post-processing around a frozen **TabFM** predictor. Other systems focus on general table-centric workflows or tool-augmented structured-data reasoning. **TabAgent** [124] solves diverse table-centric tasks within a *unified framework* with rich tool-use interfaces, structured task-solving skills, and a reliable execution harness. **TableMind** [125] and **TableMind++** [126] are tuning-based *autonomous programmatic agents* that support multi-turn planning and tool execution with lightweight LLMs. **TALON** [127] features a *planning agent* that iteratively invokes a tool agent to access and manipulate tabular data based on intermediate feedback. **MATA** [128] generates candidate answers through *diverse reasoning styles* for a given table and question, then refines or selects the optimal answer with the help of these tools. Time-series-specific workflow systems remain less common. **TimeSage**, introduced alongside the TimeSage-MT benchmark [129], combines an LLM-based agent with a time-series skill library and evaluates multi-turn analysis using verifiable targets. **TSRouter** [130] is instead a graph-based routing component that selects a modality–model pair for each query according to task context and user-specified performance–cost preferences.

Discussion. The two workflow objectives also require different evaluation criteria: predictive automation emphasizes predictive performance, pipeline validity, and resource cost, whereas structured-data reasoning emphasizes answer accuracy, execution success, and verifiability. The transition to agentic workflows introduces important trade-offs. On one hand, it raises a *reliability–efficiency trade-off*: while agents can improve adaptability to "dirty" real-world data, they also introduce substantial system overhead. In particular, LLM-based generation incurs significantly higher latency than static pipelines, and its stochastic nature can affect reproducibility. On the other hand, beyond these practical challenges, agentic workflows offer a potential integration layer for otherwise isolated models and tools. Native TSFMs and TFMs specialize in numerical prediction rather than semantic interpretation or workflow orchestration; agents may complement them through *cross-modal collaboration*. For example, a system may use a vision model to extract information, a TSFM to perform forecasting, and an LLM to synthesize a strategic report. The two directions are only beginning to converge, as many current agents still rely mainly on general LLMs, code execution, and conventional learning tools. Nevertheless, such workflows could transform structured-data models from standalone predictors into interoperable modules within a larger analytical system, supporting the vision of "language as the universal interface."

5 Future Directions

The emergence of structured foundation models provides a unified backbone for analyzing time series and tabular data, reducing the reliance on task-specific pipelines. Building on this progress, the field is now moving from *isolated universal predictors* toward *adaptive, multimodal, and composable structured-data systems* for autonomous reasoning and decision-making. We highlight several promising directions that may shape this next stage.

Towards Omni-Modal Intelligence. Future structured foundation models should move beyond modality-specific interfaces toward *native alignment* [131]: representing continuous values, schema semantics, and temporal dynamics within a shared predictive space. Current cross-modal approaches often rely on text serialization or image conversion [132], which can introduce information loss, token inefficiency, and weak numerical grounding. More direct alignment approaches [133] instead use modality-specific encoders and projectors to align heterogeneous inputs in a shared latent space while preserving structural information. More broadly, omni-modal intelligence also requires bridging the *intra-structured gap*: time series emphasize temporal order, tabular data emphasizes schema-conditioned feature interactions, and relational data introduces topology. The goal is therefore not to flatten all structured modalities into a single sequence, but to align their complementary inductive biases through a common interface.

Graphs as an emerging third structured modality. Although this survey focuses on time series and tabular data, graphs form a natural next extension of structured foundation models because they make relational topology explicit. In practice, relational databases can be viewed as graphs over entities and primary–foreign-key links, multivariate time series can be modeled as dynamic graphs over sensors or variables, and tabular instances can be connected by similarity, temporal, or

causal edges. Recent progress in graph foundation models suggests two complementary routes [134]. The first reuses tabular FMs by augmenting node features with neighborhood and structural encodings, as in **G2T-FM** [135]. For relational databases, **TabPFN-Rel** [136] follows a related route, using deep feature synthesis along foreign-key paths to construct a flat table for **TabPFN-3**'s in-context predictions. The second develops native graph or relational FMs, including **Graph-PFN**'s [137] prior-data fitting on synthetic attributed graphs and relational models such as **Griffin** [138], **Relational Transformer** [139], and **KumoRFM-2** [140]. Comparing these routes also requires reproducible evaluation: RelArena-α [136] provides a shared framework for RelBench v1's entity-level forecasting tasks, distinguishing standardized model tuning from custom system-level tuning. The central open challenge is *structural alignment*: preserving topology together with schema semantics, temporal order, and numerical precision without reverting to lossy serialization.

Unified Reasoning Engines and Scaling Laws. The next generation of models should move beyond pure "curve fitting" by supporting more explicit reasoning over structured contexts. This may involve integrating Chain-of-Thought (CoT) for time-series analysis [141], scaling test-time computation for tabular prediction [40], and supporting broader task families, from forecasting and imputation [56] to predictive table QA [142]. However, structured-data reasoning should not simply transplant linguistic CoT; it should develop schema-aware decomposition, temporal reasoning, uncertainty-aware decision rules, and table-state tracking. Reinforcement learning can further provide a training signal for reasoning and decision-making [143, 144]. At the same time, rigorous *scaling laws* are needed to quantify how performance changes with model scale [35], data properties, and compute, thereby guiding model and training design.

Dynamic Adaptation via Test-Time Training. The static "train-then-deploy" paradigm may be insufficient under real-world non-stationarity. Future systems should adopt *test-time adaptation*, where inference includes lightweight updates or adjustments based on the test input. Techniques such as Test-Time Training (TTT) [145] use the test data itself to better match the model to the current context. This ability can improve robustness under "dirty" or shifting data, treating quality changes as a form of domain shift rather than a rare anomaly.

Towards Universal Workflows and Composable Ecosystems. As the field matures, model routing is a natural systems-level direction. Given the diversity of structured tasks and data distributions, it is unlikely that a single checkpoint will be optimal everywhere. Instead, future systems may evolve into *composable ecosystems*, where a routing mechanism selects or combines suitable experts from a model zoo, balancing efficiency and performance. This macro-scale *mixture-of-experts* design can support general-purpose capability while preserving the specialization and reliability needed for high-stakes applications. Beyond routing, universal workflows must formulate tasks, compose predictors and tools, and verify outputs through standardized interfaces.

6 Conclusion

This survey presents a unified framework for Structured Data Foundation Models, bridging the domains of time series and tabular modeling. Our taxonomy systematizes approaches for *Input Encoding*, *Model Architecture*, *Pre-training Strategies*, and *Adaptation Techniques*, discussing how current methods address distinct structural constraints. Together, these developments point from *universal prediction* toward *universal workflows*, where agents can compose reusable structured-data predictors with reasoning and analytical tools. This integration is still emerging and motivates future work on *omni-modal intelligence* and *composable ecosystems*.

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